

Zichun Xia (Orey)

zichunxia78@gmail.com · orexia.com · 412-512-7363

EDUCATION

Carnegie Mellon University, School of Computer Science

Master of Science in Educational Technology and Applied Learning Science

Pittsburgh, PA

Aug 2025 – Aug 2026

Wheaton College

B.A. Studio Art, Minor in Psychology

Wheaton, IL

2019 – 2023

Relevant Courses: Web Application Development, Prompt Engineering, AI-Augmented Design, Advanced Interaction Design, Prototyping Algorithmic Experiences, Tools for Online Learning, E-Learning Design & Principles, and Evidence-Based Education Design.

AI & LEARNING EXPERIENCE

Teaching Assistant, Building with LLMs

Carnegie Mellon University × Simplilearn

Remote

Sep – Dec 2026

- Selected to support Prof. Travis Breaux as the course SME and collaborate with Simplilearn’s instructional design team on curriculum development and live delivery for an applied *Build with LLMs* program.
- Program scope builds on technical LLM topics including in-context learning, multi-stage prompting, tool use, retrieval-augmented generation (RAG), agentic workflows, and LLM evaluation; role includes supporting learner-facing curriculum materials, technical assignments, and applied activities.

Design Lead, Learnvia

Learnvia | \$55M Gates Foundation-backed AI learning nonprofit

Pittsburgh, PA

Jan – Aug 2026

- Examined when AI tutor assistance supported learner thinking versus substituted for it across 66,000+ student turns in Calculus I, operationalizing constructs including productive struggle, confusion, disengagement, and instructional effectiveness.
- Co-developed and validated an LLM-assisted behavioral coding workflow using a human-annotated reference set, iterative prompt refinement, and inter-rater reliability analysis; segmented conversations into learner-recovery trajectories and examined how scaffolding, contextual prompting, and direct answering related to subsequent learner effort.
- Translated findings into product and measurement recommendations for adaptive scaffolding, misconception diagnosis, hint sequencing, and low-confidence fallbacks, prioritizing learner recovery and independent problem solving over immediate task completion.
- Designed and developed the final research website to communicate the methodology, analytical pipeline, findings, and recommendations to project stakeholders.

AI Research Assistant, Carnegie Mellon University

17-630 Prompt Engineering Project

Pittsburgh, PA

May – Aug 2026

- Designed a 3×2 evaluation of grounded LLM responses across three question structures (generic, single Knowledge Component (KC), and multi-KC relational questions) and two workflows (direct vs. direct + verification), generating and evaluating 60 model outputs under a fixed evidence window.
- Built a verify-on-direct evaluation pipeline separating generation, source verification, and blinded scoring; evaluated faithfulness, qualification preservation, question relevance, evidence traceability, concision, and clarity.
- Found that verification improved mean response quality from 9.87 to 10.67 / 12, eliminated observed boundary violations (2 to 0), and produced the largest gain on vague questions (+1.3 pts), while also revealing an over-edit regression on already-strong responses.

AI Product Designer, Agentic Design Pipeline

n8n multi-agent orchestration for research, ideation & evaluation

Pittsburgh, PA

Mar – May 2026

- Orchestrated Input > Deep Search > User Researcher > Creative Architect > Decision Engine > Aesthetic Director > Prompt Maker, connecting research, synthesis, concept generation, evaluation, and prototype-ready output in one n8n pipeline.
- Built an A/B concept workflow that generates competing directions from shared evidence, then evaluates them against user needs, competitive context, and UX laws including Hick’s, Fitts’s, and Jakob’s Law.
- Designed structured agent handoffs and pre-mortem evaluation to reduce context drift, surface tradeoffs, and keep downstream design decisions grounded in upstream research.

AI Product Designer, ShelfSense

Pittsburgh, PA

Goodreads redesign for explainable, context-aware discovery

Jan – Feb 2026

- Reframed Goodreads discovery for readers with overloaded TBR shelves and context-driven preferences, addressing decision paralysis when saved books are abundant but next-step guidance is limited.
- Designed an explainable AI recommendation system that maps reading time, current mood, and natural-language filters to a user's TBR library, ranking titles by compatibility with match scores and personalized rationales.
- Built the product through a specification-first AI workflow, translating research into prioritized requirements and a PRD before prototyping and refining the experience in Google AI Studio.

UX Designer, City of Bridges High School

Pittsburgh, PA

Learning Media Design, Prof. Marti Louw

- Led a collaborative design inquiry with teachers and students to increase engagement of quieter learners in an Art History course through spatial and interaction design.
- Designed and prototyped a dual-modality system (“Emotion Aura Map” physical installation + “Learning Art Gallery” dashboard) to make silent participation visible and actionable for instructors.
- Conducted storyboard speed dating and think-aloud testing with high school learners to iterate high-fidelity prototypes grounded in learning science and student agency.

RESEARCH & TEACHING EXPERIENCE

Research Assistant, Materialities Lab and Sound Studies Group

Shenzhen, China

Southern University of Science and Technology, School of Design

Mar – Jul 2024

- Built the *A Grain Of* dataset by collecting 160+ videos, audio recordings, and photographs and processing them into point-cloud archives; applied CNNs to detect anthropogenic traces.
- 3D-scanned 210+ material samples and produced 25 remeshed, resin-printed models combining physical forms with audio spectrograms.
- Prototyped custom Arduino/ESP32 hardware and curated a 2m³ interactive installation for 400 visitors at Shenzhen Design Week, integrating haptics, spatial audio, and projection mapping.
- Developed an interactive website for the *A Grain Of* archive, featuring a digital archive, dynamic data visualization, multimedia integration, and user engagement through JavaScript, HTML, and CSS.

Design Project Assistant, Eco-centric Future Lab

Austin, TX

University of Texas at Austin, mentored by Prof. Jiabao Li

Dec 2023 – Mar 2024

- Collaborated with Jiabao Li and Ben Evanson on *Carbon Farm*, featured in the exhibition *Progenitorial Hysteresis*.
- Supported the fabrication and deployment of the interactive carbon-capture installation, which transformed participants' breath and speech into inputs for algae cultivation.
- Assisted with physical prototyping, interaction setup, and the on-site participant experience.

Teaching Assistant, DS-366 Composing Experience

Shenzhen, China

SUSTech, under Prof. Marcel Zaes Sagesser

Mar – Jun 2024

- Lectured a DS-366 class of twenty-six students on fundamentals of Adobe After Effects.
- Provided individualized guidance to students in the completion of projects, enabling them to create visually compelling and animated design experiences for local corporate clients and designated prompts.

Teaching Assistant, EDUC 135 School and Society

Wheaton, IL

Wheaton College, under Prof. Mark Jonas

Sep – Dec 2021

- Facilitated semester-long peer-mentoring drop-in sessions to support 14 students in enhancing the clarity, coherence, and logical structure of their mid-term and final philosophical argumentative essays.

SKILLS

Curriculum & Learning Design: Curriculum Development, Backward Design, Scaffolding, Knowledge Component Mapping, Cognitive Task Analysis (CTA), Assessment Design, SME Collaboration, KLI Framework

AI & Prototyping: Claude, Cursor, n8n, Prompt Engineering, Prompt Chaining, Multi-Agent Workflow Orchestration, LLM Evaluation, API Integration, Python, Pandas, React

Research & Evaluation: Experimental Design, Qualitative Coding, Inter-Rater Reliability, Conversation Analysis, Mixed-Methods Analysis, Rubric-Based Evaluation, Think-Aloud Studies

Product Design: Figma, Framer, Vercel, Interaction Design, Design Systems, User Flows, Rapid Prototyping